

Event Modelling

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1 Wallet Lifecycle Events:

1.1 WalletCreated

What it is

Represents the creation of a new wallet in the system. This event establishes the existence of a wallet and marks the starting point of the wallet's lifecycle. No other lifecycle or financial events are valid before this event occurs.

When it happens

This event is emitted when a create-wallet command is successfully validated and accepted by the Command Service. It occurs only once for a given wallet and before any other wallet-related operation.

What data it carries

The event carries the minimum information required to uniquely identify the wallet and audit its creation:

- Wallet identifier generated by the system
- Client identifier representing the calling system
- Client request identifier used for idempotency
- Initial wallet status
- Initial balance
- Timestamp of creation

What rules it enforces (Invariants)

- A wallet must not already exist with the same wallet identifier.
- A wallet can be created only once in its lifetime.
- No credit, debit, reversal, or lifecycle transition is allowed before this event occurs.
- By default initial status of wallet would be active.

1.2 WalletActivated

What it is

Represents the activation of an existing wallet, indicating that the wallet is now allowed to perform financial operations such as credit and debit. This event transitions the wallet from a non-operational state to an operational state.

When it happens

This event is emitted when an activate-wallet command is successfully validated and accepted by the system for an existing wallet that is eligible for activation.

What data it carries

The event contains the following information:

- Wallet identifier of the wallet being activated
- Client identifier representing the calling system
- Client request identifier used for idempotency
- New wallet status (ACTIVE)
- Timestamp indicating when the activation occurred

What rules it enforces (Invariants)

- A wallet must not already be in the active state.
- A wallet that is closed must not be activated again.
- Financial operations (credit/debit) are allowed only after this event has occurred.

1.3 WalletSuspended

What it is

Represents the temporary suspension of an existing wallet. A suspended wallet remains present in the system but is not allowed to perform any financial operations until it is reactivated or closed.

When it happens

This event is emitted when a suspend-wallet command is successfully validated and accepted for an existing, active wallet.

What data it carries

The event contains the following information:

- Wallet identifier of the wallet being suspended
- Client identifier representing the calling system
- Client request identifier used for idempotency
- New wallet status (SUSPENDED)
- Timestamp indicating when the suspension occurred

What rules it enforces (Invariants)

- The wallet must currently be in the ACTIVE state.
- A wallet that is already SUSPENDED must not be suspended again.
- A wallet that is CLOSED must not be suspended.
- While suspended, the wallet must not allow credit, debit, or reversal operations.

1.4 WalletClosed

What it is

Represents the permanent closure of a wallet, once closed the wallet is considered terminated and can no longer participate in any lifecycle or financial operations. This event marks the end of the wallet's lifecycle.

When it happens

This event is emitted when a close-wallet command is successfully validated and accepted for an existing wallet that is eligible for closure.

What data it carries

The event contains the following information:

- Wallet identifier of the wallet being closed
- Client identifier representing the calling system
- Client request identifier used for idempotency
- New wallet status (CLOSED)
- Timestamp indicating when the closure occurred

What rules it enforces (Invariants)

- A wallet that is already CLOSED must not be closed again.
- Once closed, the wallet must never transition to any other state.
- A closed wallet must not any kind of operations.

2 Wallet Operations:

2.1 WalletCredited

What it is

Represents the addition of funds to an existing wallet. This event increases the wallet's balance and records a permanent financial fact that the specified amount was successfully credited.

When it happens

This event is emitted when a credit-wallet command is successfully validated and accepted for an existing, active wallet.

What data it carries

The event contains the following information:

- Wallet identifier of the wallet being credited
- Credited amount
- Client identifier representing the calling system
- Client request identifier used for idempotency
- Updated wallet balance after the credit

- Timestamp indicating when the credit was recorded

What rules it enforces (Invariants)

- The wallet must currently be in the ACTIVE state.
- The credited amount must be greater than zero.

2.2 WalletDebited

What it is

Represents the successful deduction of funds from an existing wallet. This event permanently records that a specified amount has been debited from the wallet and reduces the wallet's balance accordingly.

When it happens

This event is emitted when a debit-wallet command is successfully validated and accepted for an existing, active wallet.

What data it carries

The event contains the following information:

- Wallet identifier of the wallet being debited
- Debited amount
- Client identifier representing the calling system
- Client request identifier used for idempotency
- Updated wallet balance after the debit
- Timestamp indicating when the debit was recorded

What rules it enforces (Invariants)

- The wallet must be in the ACTIVE state.
- The debited amount must be greater than zero.
- The wallet balance must be greater than or equal to the debited amount.

3 Reversal Operations

3.1 CreditReversed

What it is

Represents the reversal of a previously recorded credit operation.

When it happens

This event is emitted when a reverse-credit command is successfully validated and accepted for an existing wallet, referencing a specific prior WalletCredited event.

What data it carries

The event contains the following information:

- Wallet identifier of the wallet
- Reference to the original credited event being reversed
- Reversed amount
- Client identifier representing the calling system
- Client request identifier used for idempotency
- Updated wallet balance after the reversal
- Timestamp indicating when the reversal was recorded

What rules it enforces (Invariants)

- The referenced WalletCredited event must exist.
- The wallet must not be in the CLOSED state at the time of reversal.
- The reversal amount must exactly match the original credited amount.

3.2 DebitReversed

What it is

Represents the reversal of a previously recorded debit operation.

When it happens

This event is emitted when a reverse-debit command is successfully validated and accepted for an existing wallet, referencing a specific prior WalletDebited.

What data it carries

The event contains the following information:

- Wallet identifier of the wallet
- Reference to the original credited event being reversed
- Reversed amount
- Client identifier representing the calling system
- Client request identifier used for idempotency
- Updated wallet balance after the reversal
- Timestamp indicating when the reversal was recorded

What rules it enforces (Invariants)

- The referenced WalletDebited event must exist.
- The wallet must not be in the CLOSED state at the time of reversal.
- The reversal amount must exactly match the original debited amount.

Common Invariants for all events:

- Wallet must exist.
- Any event must **not be applied more than once** for the **same (clientId, clientRequestId)**.